Chassis Material Menu

What Should We Build the Frame From?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different building systems and ways to solve the same design problem.
- **[Red: Critical Thinking & Problem Solving]** We compare material trade-offs before building.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and available materials.
- **[Orange: Cultural Understanding & Civic Engagement]** We use shared and borrowed materials responsibly.
- **[Blue: Effective Communication]** We explain our material choice in the engineering portfolio.

Decision Being Made

Your team is choosing what materials or building system to use for the robot chassis. The chassis shape matters, but the material also matters. Different systems have different strengths, fasteners, hole patterns, sizes, connectors, and tools.

Your team may not have unlimited access to every part or every length. This is normal. Real engineering teams design with the materials, money, time, and tools they actually have.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Physics and Space Planning**
- **Borrowed Equipment and Community Trust**
- materials available in the classroom,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

The frame material affects:

- how strong the chassis is,
- how easy it is to build,
- how easy it is to add attachments,
- how much space the frame uses,
- how easy the robot is to repair,
- how well parts from different systems fit together,

- how much time the team spends on the base instead of scoring mechanisms,
- and how responsibly the team uses shared or borrowed materials.

There is no perfect material choice. Your team should choose the material that best fits your design, skill level, available parts, and time.

How to Use This Menu

- Review the material options.
- Compare strength, ease of use, time, and attachment options.
- Check what materials are actually available.
- Decide whether your team should build a new chassis, modify an existing chassis, or recycle a known chassis.
- Record the choice, trade-off, and evidence plan in the engineering portfolio.

Menu Options

Option A: goBILDA U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: goBILDA U-channel is a rigid metal channel with many holes for attaching brackets, motors, wheels, shafts, and other parts.

What it helps: It is strong, precise, and designed for robotics. It works well for sturdy frames and reliable mounting.

Best for: Teams that want a strong chassis with many attachment points.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: goBILDA channel, compatible screws, nuts, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Specific lengths may not be available. Some parts may not line up cleanly with REV or Tetrix parts. Working inside channel can become hard after the robot is assembled.

Evidence we might watch: Does the frame stay rigid? Are attachments easy to add? Can we reach screws, motors, battery, and electronics?

Option B: REV U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: REV U-channel is a metal channel used in the REV building system. It provides structure and attachment points.

What it helps: It can create strong frame members and may work well with REV brackets, plates, shafts, motors, and electronics.

Best for: Teams using mostly REV parts or needing a simple channel-based frame.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material / tooling needs: REV channel, compatible fasteners, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Hole patterns and dimensions may not match goBILDA or Tetrix parts perfectly. Teams may need adapters or creative brackets.

Evidence we might watch: Can we attach wheels, motors, and mechanisms securely? Do parts line up well enough without forcing them?

Option C: REV 15 mm Extrusion

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: REV 15 mm extrusion is a smaller beam-like structure that allows parts to be placed in many positions along the rail.

What it helps: Extrusion can be flexible for mounting because parts can often slide or be placed continuously instead of only at fixed holes. It is compact and useful for adjustable structures.

Best for: Teams that need flexible attachment points, small structures, supports, or adjustable mounts.

Ease of use: Medium to hard

Likelihood of success: Medium

Time to implement: Moderate to slow

Material / tooling needs: REV extrusion, brackets, corner connectors, compatible screws, nuts, and tools.

Possible trade-offs: It may be less obvious for beginners to assemble. It may require special connectors. A full chassis made only from extrusion may need careful bracing to stay square and rigid.

Evidence we might watch: Can we attach parts where we want? Does the frame stay square? Does anything slide, loosen, or twist during driving?

Option D: Tetrix Channel or Mixed Channel Parts

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Tetrix and other channel parts may be available. Teams may combine parts from different systems when the exact preferred material is not available.

What it helps: Mixed systems can make use of available materials and may solve a parts shortage.

Best for: Teams that are flexible, willing to test fit, and able to use brackets or plates to connect parts from different systems.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Tetrix channel, compatible fasteners, adapters, brackets, plates, and possibly parts from multiple systems.

Possible trade-offs: Different systems may not line up cleanly. Holes, screws, shafts, hubs, and spacing may not match. Teams may spend extra time solving connection problems.

Evidence we might watch: Are connections secure? Are parts being forced together? Does the frame stay aligned after driving?

Option E: Hybrid Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team intentionally combines systems. For example, the main chassis might use goBILDA channel, while smaller adjustable mounts use REV extrusion.

What it helps: A hybrid frame can use the strengths of different systems. Strong channel can support the base, while extrusion or brackets can help with adjustable mechanisms.

Best for: Teams that have a clear reason for mixing systems and enough patience to manage fit issues.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Parts from multiple systems, adapter plates or brackets, compatible fasteners, and careful test fitting.

Possible trade-offs: Mixing systems can create alignment, spacing, and fastener problems. It may take more time than using one system.

Evidence we might watch: Does each system solve a specific problem? Are mixed connections strong and repairable?

Option F: Recycle or Modify a Known Chassis

Graduate Expectation Connection: [Green: Personal Development], [Orange: Cultural Understanding & Civic Engagement], and [Red: Critical Thinking & Problem Solving]

What it means: Instead of building a brand-new chassis from scratch, the team uses an existing chassis, known base, or partially built robot frame and modifies it for the challenge.

What it helps: This can save time, reduce risk, and give the team more time for driving, programming, shooter design, hopper/trigger work, testing, and portfolio development.

Best for: Teams that want a reliable starting point, have limited build time, or need to focus on scoring mechanisms and controls.

Ease of use: Easy to medium

Likelihood of success: High if the chassis is in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, repair tools, possible replacement screws, wheels, hubs, motors, or brackets.

Possible trade-offs: The team may have less ownership over the original design. The existing chassis may not perfectly fit the team's strategy. Old choices may limit where attachments, battery, electronics, or game-piece paths can go.

Evidence we might watch: Does the existing chassis drive reliably? What design limits does it create? How much time did it save? What did the team modify?

Comparison Table

Option	Best Feature	Main Risk	Good For
goBILDA U-channel	Strong and precise	Specific lengths/parts may be limited	Strong chassis base
REV U-channel	Works well with REV ecosystem	May not align with other systems	REV-based builds
REV 15 mm extrusion	Flexible placement	Needs careful connectors/bracing	Adjustable mounts and compact structures
Tetrix / mixed channel	Uses available parts	Fit and fastener problems	Teams solving material shortages
Hybrid frame	Uses strengths of multiple systems	More complex connections	Teams with a clear mixed-system plan
Recycled known chassis	Saves time and reduces risk	Less original design control	Teams prioritizing mechanisms, code, and testing

Materials Availability Check

Before finalizing your material choice, check:

- Do we have enough pieces in the lengths we need?
- Do we have the correct fasteners?
- Do we have brackets or connectors that fit?
- Do we have the correct hubs, shafts, spacers, and wheels?
- Can we attach motors securely?
- Can we reach screws later for repair?
- Are we using borrowed materials responsibly?
- Do we need to request a limited part?

Material Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Main frame material				
Connector/bracket plan				
Mixed systems?				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Recycled chassis?				

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis material or existing chassis and explain why it is good enough to start building.
Should Do	Explain the main trade-off of the material choice and check whether the needed parts are available.
Could Do	Compare two material systems or compare new-build vs. recycled chassis using time, reliability, repair access, and future mechanism space.

SLD Connection

Main SLD Prompt

Should our team build a new chassis from preferred materials, adapt to the materials available, or recycle a known chassis to save time?

Supporting Questions

- What material gives us the best chance of a reliable driving base?
- What material gives us the most flexibility for future attachments?
- What material is actually available?
- How much time should we spend on the chassis before moving to mechanisms?
- When is recycling a known chassis a smart engineering decision?
- What do we lose if we do not build the chassis from scratch?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis material or existing chassis,
- reason for the choice,
- material trade-off statement,
- parts availability notes,
- sketch or photo of planned frame,
- and evidence the team will watch during testing.

Reflection

Our chassis material choice is:

We chose this because:

The main trade-off we accepted is:

One material constraint we need to manage is: